

CS1 Week 11

Empty

Frequency-domain Specifications, Loop shaping



Recap Last Week

Quiz 1

What does a single point in the polar plot of a system $G(j\omega)$ represent?

A. The open-loop gain k at frequency ω

B. The complex value $G(j\omega)$, consisting of magnitude and phase

C. Only the phase of $G(j\omega)$, plotted as an angle on the imaginary axis

D. Only the magnitude of $G(j\omega)$, plotted as a radius

Quiz 2

Which of the following statements are correct?

- A.** With Nyquist we can determine closed-loop stability only by looking at the open-loop
- B.** $N = Z - P$, with N = encirclements about $-1/k$; Z = open-loop zeros; P = open-loop poles
- C.** $N = Z - P$, with N = encirclements about $-1/k$; Z = closed-loop poles; P = open-loop poles
- D.** Nyquist is a more robust way to determine stability of a closed-loop system

Quiz 3

Which of the following statements about Nyquist plots are correct?

- A.** They are symmetric w.r.t. the real axis
- B.** The frequency is implicit in the Nyquist plot
- C.** We can add Nyquist Plots of different TFs on top of each other
- D.** They start at $\omega = \infty$ and end at $\omega = 0$

Course Schedule

Subject		Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control	8
	Frequency response, Bode plots	9
	Polar plots, The Nyquist condition	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

Today

1. Gain & Phase Margin
2. Frequency-Domain Specifications
3. Loop Shaping

1. Gain & Phase Margin

$$N = Z - P$$

Gain & Phase Margin

- These margins are only valid for open-loop stable systems! ($P = 0$) → for stability we need $N = 0$
- Phase & Gain Margin define "how close" we are to closed loop instability

(i.e. encircling the -1 point)

1. Gain Margin: $g_m = \frac{1}{|L(j\omega_{pc})|}$

How much can we change the gain at -180° until we hit -1 ?

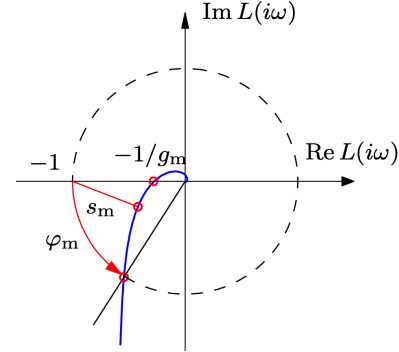
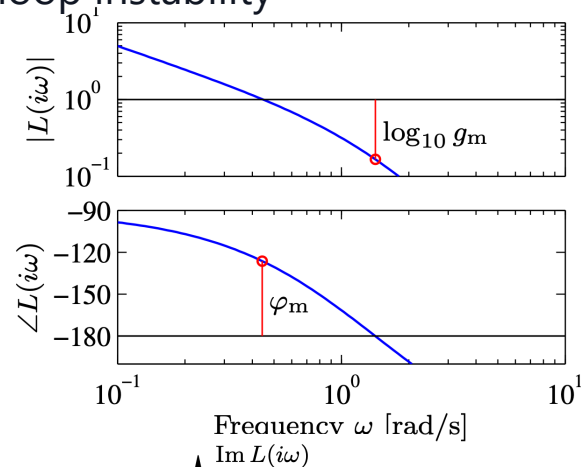
2. Phase Margin: $\varphi_m = 180^\circ + \angle L(j\omega_c)$

How much can we change the phase at 0dB until we reach -1 ?

3. Bandwidth:

Frequency where magnitude equals $\frac{1}{\sqrt{2}} = -3dB \approx \omega_{gc}$

measures how fast CL-System can react



Exam Problem

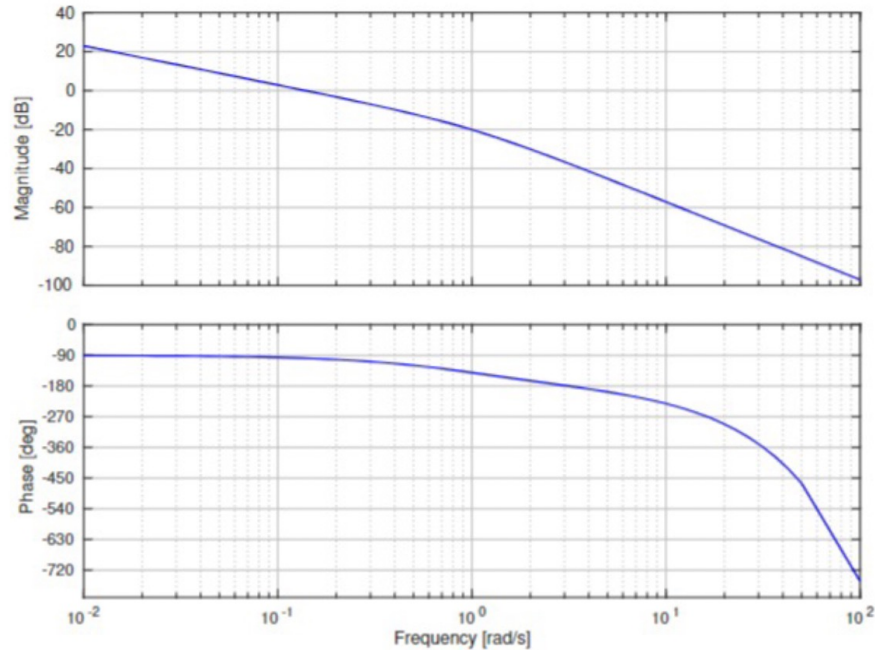
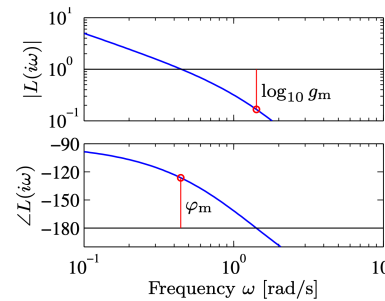


Figure 14: Bode plot of L .



1. Gain Margin:

How much can we change the gain at -180° until we hit -1 ?

2. Phase Margin:

How much can we change the phase at 0dB until we reach -1 ?

Q29 (0.5 Points) Mark the correct answer.

The phase margin PM for closed-loop stability is best described by,

☐ A $PM \approx -83^\circ$

☐ C $PM \approx 47^\circ$

☐ B $PM \approx 80^\circ$

☐ D $PM \approx -35^\circ$

Q30 (0.5 Points) Mark the correct answer.

The gain margin GM for closed-loop stability is best described by,

☐ A $GM \approx -37$ dB

☐ C $GM \approx 37$ dB

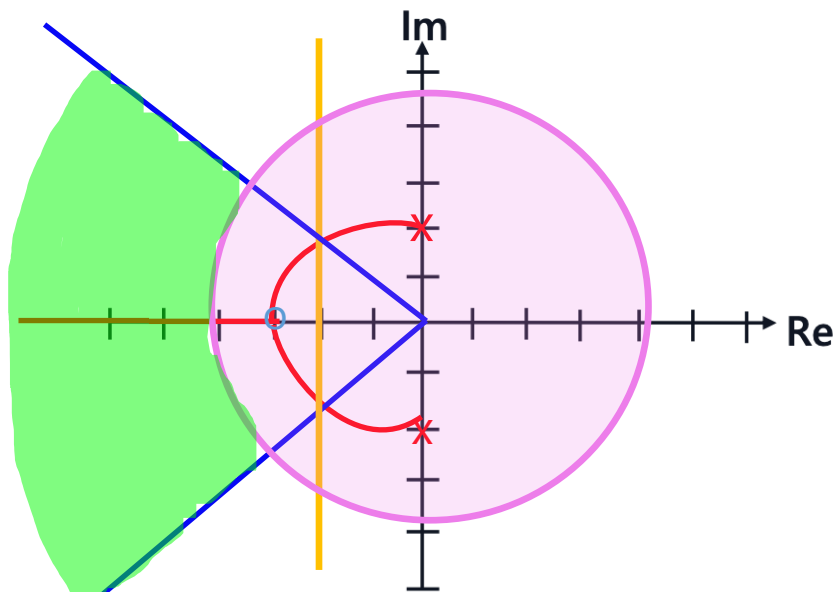
☐ B $GM \approx 82$ dB

☐ D $GM \approx -21$ dB

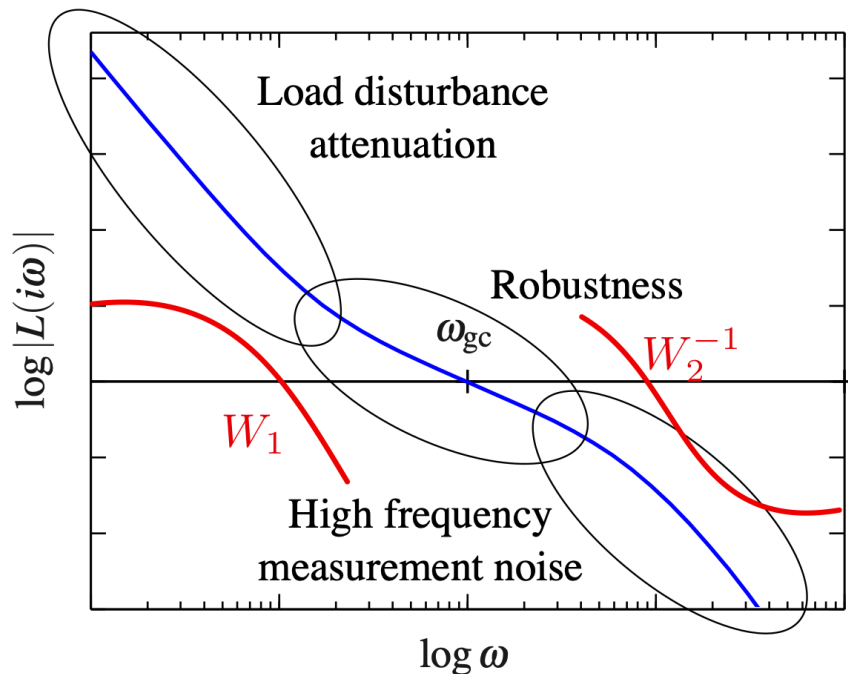
2. Frequency-domain Specifications

Overview

Three weeks ago, we looked at the **time-domain specifications** which looked like this:

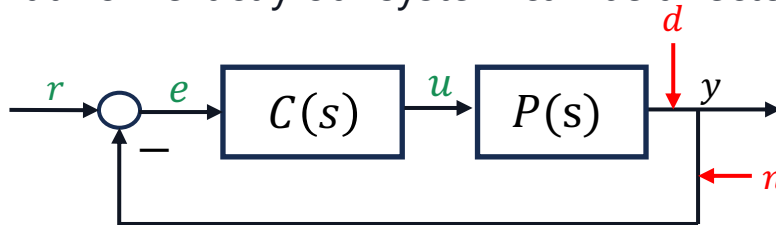


Today, we're going to look at the **frequency-domain specifications** which look like this:



Noise & Disturbance

Let's take a closer look at how exactly our system can be affected:



The effect of d and n on the system can be seen from the corresponding transfer functions:

- $d \rightarrow e$:
$$S(s) = \frac{1}{1+C(s)P(s)}$$
- $n \rightarrow e$:
$$T(s) = \frac{C(s)P(s)}{1+C(s)P(s)}$$

- The effect of both n and d on e should be as small as possible
- That means, both $S(s)$ and $T(s)$ should be as small as possible

There's one small issue...

$$\bullet \quad d \rightarrow e:$$

$$S(s) = \frac{1}{1+C(s)P(s)}$$

$$\bullet \quad n \rightarrow e:$$

$$T(s) = \frac{C(s)P(s)}{1+C(s)P(s)}$$

(Complementary) Sensitivity Functions

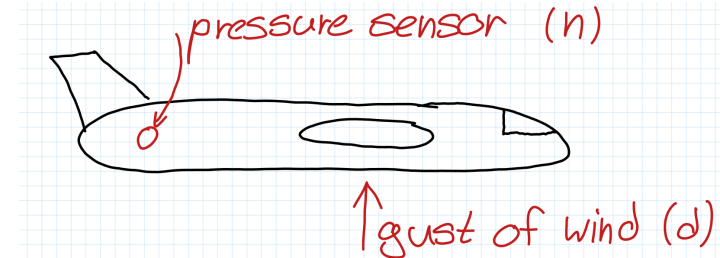
$$S(s) + T(s) = \frac{1}{1+C(s)P(s)} + \frac{C(s)P(s)}{1+C(s)P(s)} = 1$$

→ They always add up to one!

- This means that we can't cancel both noise and disturbances...

Thankfully, they have a property we can make use of:

- Noise usually occur at high frequencies
- Disturbances usually occur at low frequencies



Frequency-domain Specifications

So the workaround is to require:

1) $|S(s)| \ll 1$ for small ω
→ disturbance rejection

2) $|T(s)| \ll 1$ for large ω
→ noise rejection

How can we guarantee these constraints?

$$1) S(s) = \frac{1}{1+L(s)} \Rightarrow S(s) \text{ small for small } \omega$$

We enforce the inequality using a constraint function $W_1(s)$: $|S(s)||W_1(s)| < 1$

for some $W_1(s)$ that is large at low ω

The equation can be rewritten as:

$$\approx |L(s)| > |W_1(s)| \left| \frac{1}{1+L(s)} \right| |W_1(s)| < 1 \\ \Rightarrow |1+L(s)| > |W_1(s)|$$

$$2) T(s) = \frac{L(s)}{1+L(s)} \Rightarrow T(s) \text{ small at high } \omega$$

Again using a constraint function $W_2(s)$:

$$|T(s)||W_2(s)| < 1$$

for some $W_2(s)$ that is large at high ω

The equation can be rewritten as:

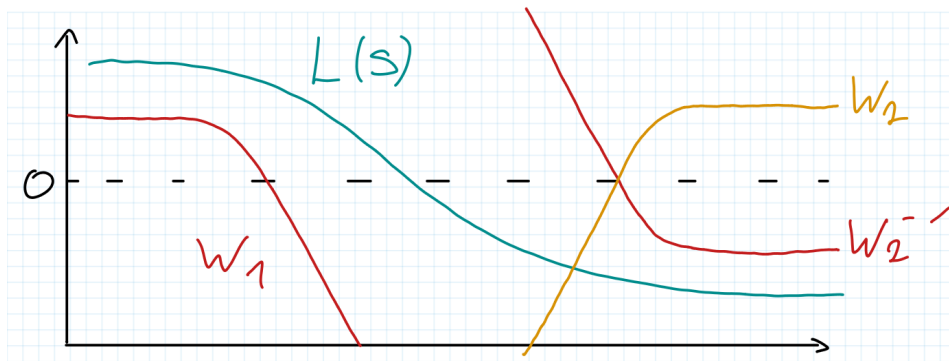
$$\left| \frac{L(s)}{1+L(s)} \right| |W_2(s)| < 1 \Rightarrow |L(s)| |W_2(s)| < 1 \\ \approx |L(s)| < |W_2(s)|^{-1}$$

Visualization

$$|L(s)| > |W_1(s)|$$

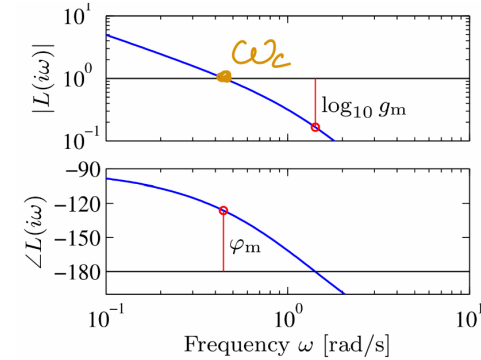
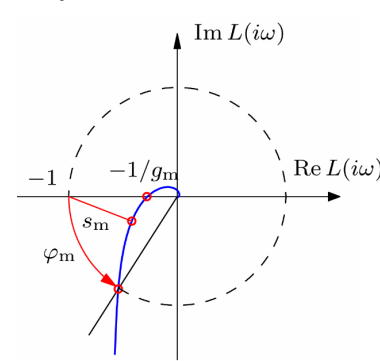
$$|L(s)| < |W_2(s)|^{-1}$$

What exactly are these $W(s)$? :



- The W 's are like obstacles for the open-loop TF
 - There's *one more important condition* for the closed-loop: $T(s) = \frac{L(s)}{1+L(s)}$
- system becomes unstable for $L(s) = -1$
- We could also see this in the Bode-plot!

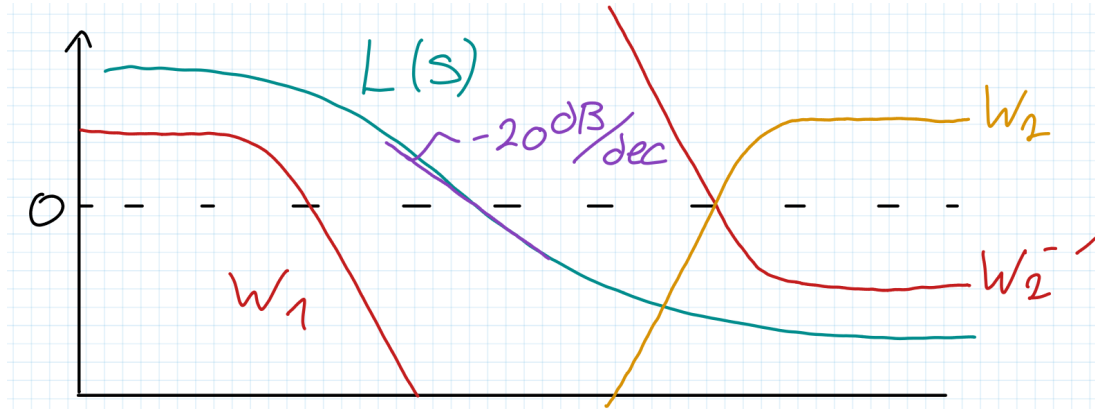
Bode Plot



Another condition:

The slope of the magnitude plot is related to the phase!

- To ensure enough phase margin, the slope of the magnitude plot at the crossover frequency ω_c should be not steeper -20dB/dec



Dynamic Compensator

- All of these conditions for stability and performance make sense
 - But our plant $P(s)$ just behaves however it behaves, we cannot change the physics
- How do we ensure that the open loop $L(s)$ looks how we want?

⇒ Choose a controller $C(s)$ s.t. $L(s)$ does what we want!

Example: Given $P(s) = 0.01 \frac{(s+10)^2}{s(s^2+2s+2)}$

Find a $C(s)$ such that the following conditions are respected:

- $W_1 = 100\text{dB}$ for $\omega < 0.1\text{rad/s}$
- $W_2^{-1} = 0.05\text{dB}$ for $\omega > 100\text{rad/s}$
- $\omega_c = 10\text{rad/s}$
- 45° phase margin at ω_c

Dynamic Compensator

- A good open-loop suggestion would be:

$$L_{\text{des}} = 10 \cdot \frac{1}{s \left(\frac{s}{50} + 1 \right)}$$

- Now we reverse engineer:

$$C(s) = \frac{L_{\text{des}}(s)}{P(s)} = 10 \cdot \frac{s^2 + 2s + 2}{(s/50 + 1)(s/10 + 1)^2}$$

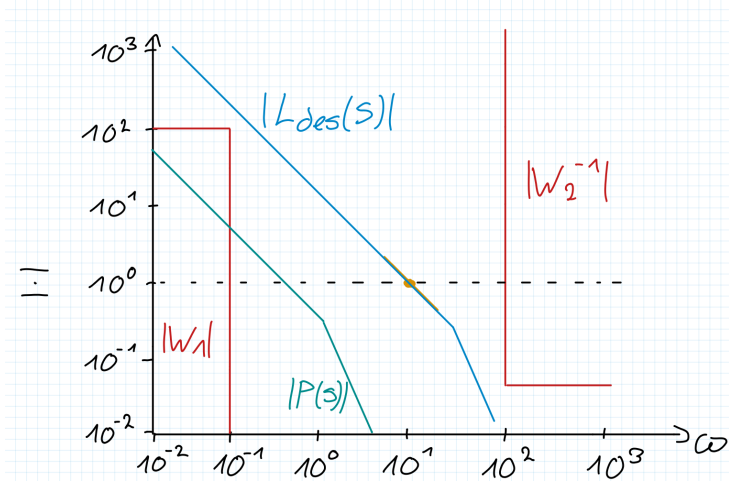
What we just did was a “brute force” approach...

- $C(s)$ may be unnecessarily complicated
- **Only works for open-loop stable, minimum-phase systems!**

otherwise, we would cancel unstable poles

$$P(s) = 0.01 \frac{(s + 10)^2}{s(s^2 + 2s + 2)}$$

- $W_1 = 100\text{dB}$ for $\omega < 0.1\text{rad/s}$
- $W_2^{-1} = 0.05\text{dB}$ for $\omega > 100\text{rad/s}$
- $\omega_c = 10\text{rad/s}$
- 45° phase margin at ω_c



A more elegant approach is called Loop Shaping...

Exam Question

Box 3: Questions 31, 32

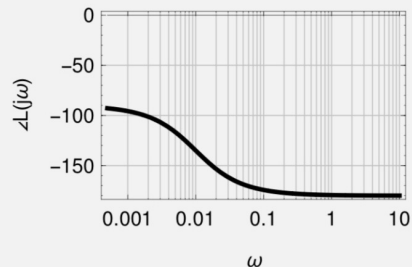
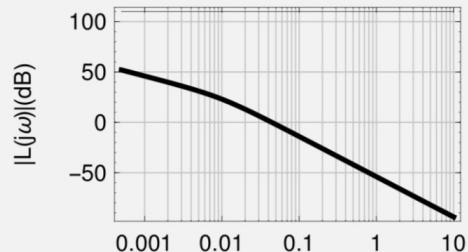
You are faced with the task of designing a controller $C(s)$ for the plant:

$$P(s) = \frac{1}{500 \left(s^2 + \frac{s}{100} \right)}$$

and decide to use loop-shaping to solve the problem. You face the following constraints:

- Disturbances at frequencies $\omega \leq 0.01$ should be reduced by at least $60dB$.
- Noise which occurs at frequencies $\omega \geq 100$ should be attenuated by at least $100dB$.

The Bode plot of $P(s)$ is displayed below:



dB	decimal
20	10
40	100
60	1 000
80	10 000
100	100 000

$$|L(s)| > |W_1(s)|$$

$$|L(s)| < |W_2(s)|^{-1}$$

Question 31 Choose the correct answer. (1 Point)

How can the disturbance rejection and noise reduction specifications be represented in the Bode plot?

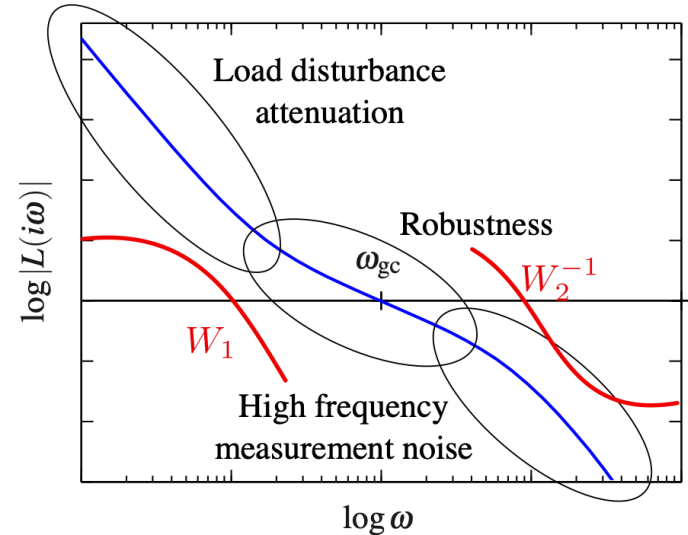
- ☐ A $|L(j\omega)| \approx 60dB$ for $\omega < 0.01$ and $|L(j\omega)| > -100dB$ for $\omega > 100$
- ☐ B $|L(j\omega)| > 60dB$ for $\omega < 0.01$ and $|L(j\omega)| > -100dB$ for $\omega > 100$
- ☐ C $|L(j\omega)| > 60dB$ for $\omega < 0.01$ and $|L(j\omega)| < -100dB$ for $\omega > 100$
- ☐ D $|L(j\omega)| < 60dB$ for $\omega < 0.01$ and $|L(j\omega)| > -100dB$ for $\omega > 100$
- ☐ E $|L(j\omega)| < 60dB$ for $\omega < 0.01$ and $|L(j\omega)| < -100dB$ for $\omega > 100$

3. Loop Shaping

Loop Shaping

A better way to design our Compensator $C(s)$ is to use the following elements, s.t. $L(s)$ passes the obstacle course:

- Gain: k
- Integrator: $\frac{1}{s}$
- Lead: $\frac{s/a+1}{s/b+1}, 0 < a < b$
- Lag: $\frac{s/a+1}{s/b+1}, 0 < b < a$



⇒ We can then add each element in series how we want, i.e. $C(s) = C_1(s) \cdot C_2(s) \cdots C_n(s)$

⇒ But what is the effect of Lead/Lag-elements?

Lead-Element

- In essence, Lead-Lag elements are a combination of a stable pole & mm. phase zero
- We use them to manipulate the frequency response

$$C_{lead} = \frac{s/a + 1}{s/b + 1}, 0 < a < b$$

Effects:

- The magnitude at high frequencies is **increased** by b/a . Low frequencies are not affected.
- The slope of the plot is increased by $+20\text{dB/dec}$ between a and b .
- The phase is **increased** around $\sqrt{ab}$ by up to 90° . The maximum phase increase is:

$$\varphi_{\max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90^\circ$$

Magnitude
response

Phase
response

Bode Plot of a Lead Element

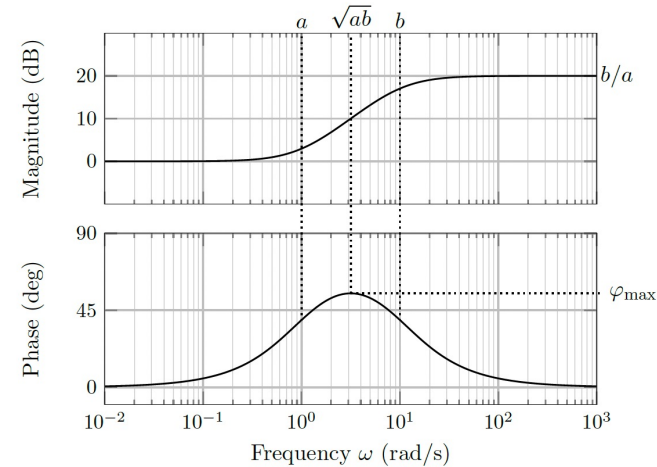


Figure 7.14: Bode Plot of a lead element with $a = 1$ and $b = 10$

Lag-Element

- A lag compensator is in many ways an inverse of the lead compensator

$$C_{lag} = \frac{s/a + 1}{s/b + 1}, 0 < b < a$$

Effects:

- The magnitude at high frequencies is **decreased** by b/a . Low frequencies are not affected.
- The slope of the plot is decreased by -20 dB/dec between b and a .
- The phase is **decreased** around $\sqrt{ab}$ by up to 90° . The maximum phase decrease is:

$$\varphi_{\max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90^\circ$$

Magnitude response

Phase response

Bode Plot of a Lag Element

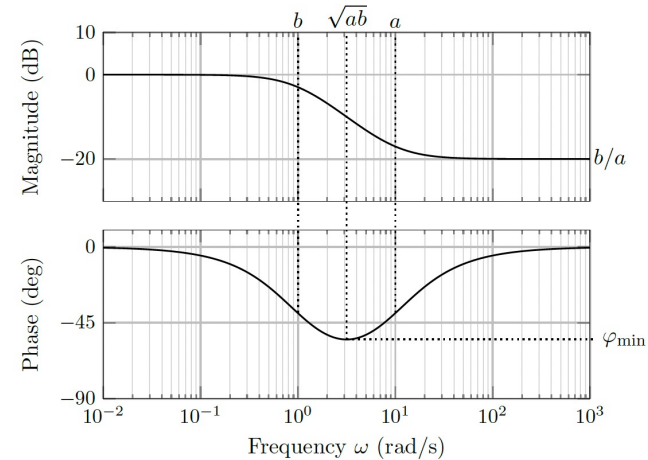


Figure 7.15: Bode Plot of a lag element with $b = 1$ and $a = 10$

Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5 \frac{(1 - \frac{s}{200})}{(1 + \frac{s}{20})(1 + \frac{s}{4})}$$

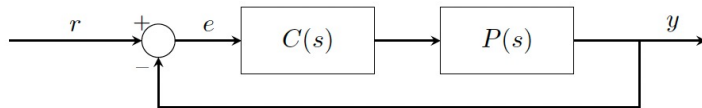


Figure 3: Standard feedback architecture.

In order to design the compensator, the problem has been split into the following subtasks:

3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45° .

a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.

Hints 1 & 2:

- nmn-phase zero limits performance
 - restrict how ω_{gc} can be set
 - rise time linked to bandwidth
 - common rule $\omega_{gc} < 0.5z^+$
- higher $\omega_{gc} \rightarrow$ faster rise time

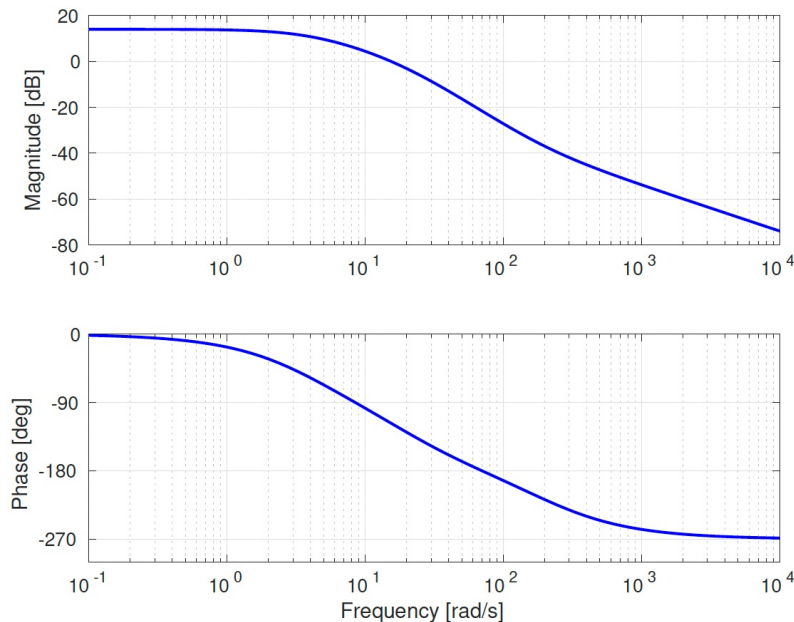


Figure 5: Bode plot for $P(s)$.

Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5 \frac{(1 - \frac{s}{200})}{(1 + \frac{s}{20})(1 + \frac{s}{4})}$$

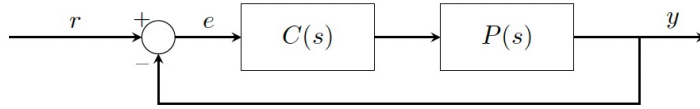


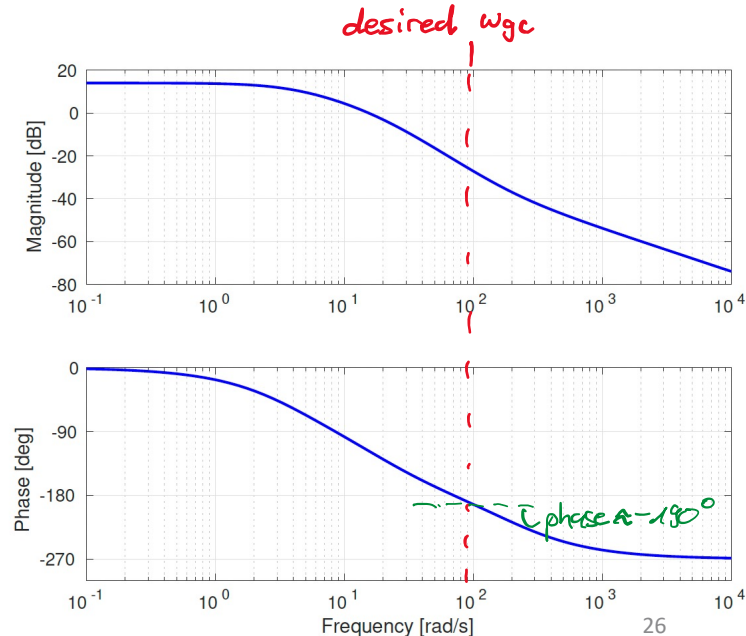
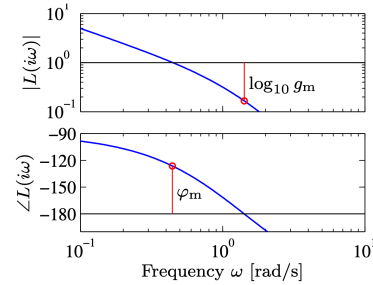
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- Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.
- Determine the form of compensator required to achieve the desired crossover frequency and phase margin.

$$\omega_c = 10^2$$



Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5 \frac{(1 - \frac{s}{200})}{(1 + \frac{s}{20})(1 + \frac{s}{4})}$$

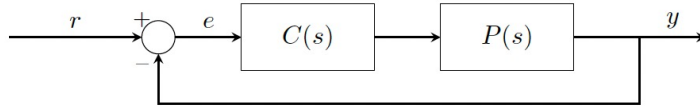


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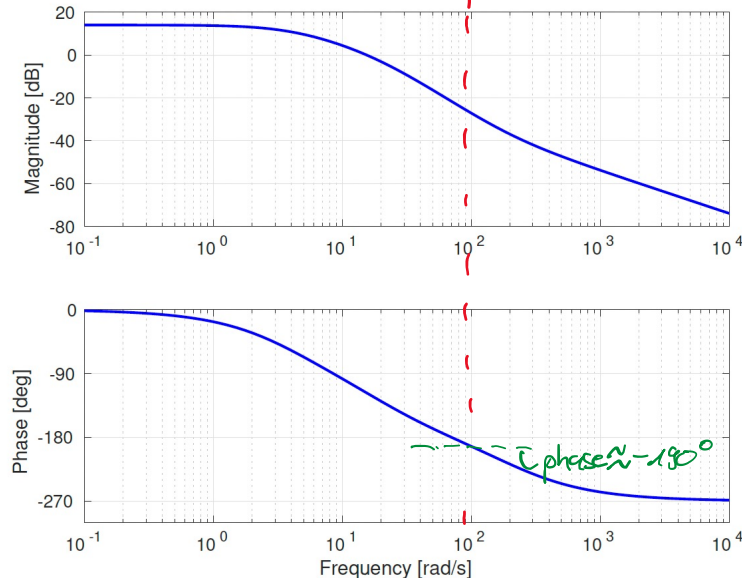
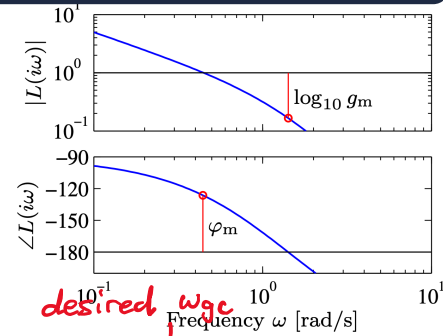
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3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45° .

- Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.
- Determine the form of compensator required to achieve the desired crossover frequency and phase margin. **Lead Compensator**
- Determine the desired phase lead/lag that the compensator must provide at the crossover frequency ω_c .

Hints:

Use your result from (a) to determine the phase of $P(j\omega)$ at $\omega = \omega_c$ and hence, the desired phase shift.



Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5 \frac{(1 - \frac{s}{200})}{(1 + \frac{s}{20})(1 + \frac{s}{4})}$$

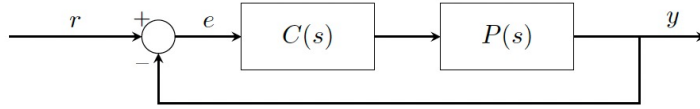


Figure 3: Standard feedback architecture.

In order to design the compensator, the problem has been split into the following subtasks:

3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45° .

- a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system. $\omega_c = 10^2$
- b) Determine the form of compensator required to achieve the desired crossover frequency and phase margin. Lead Compensator
- c) Determine the desired phase lead/lag that the compensator must provide at the crossover frequency ω_c . 57.97°
- d) At what point do we want the maximum phase shift of our compensator to occur? Use this to formulate a requirement on the parameters a and b. Combine it with your result from (c) to determine a and b.

Hints:

The maximum phase shift of a lead compensator occurs at $\omega = \sqrt{ab}$

$$\varphi_{\max} = 2 \arctan \left(\sqrt{\frac{b}{a}} \right) - 90^\circ$$

Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5 \frac{(1 - \frac{s}{200})}{(1 + \frac{s}{20})(1 + \frac{s}{4})}$$

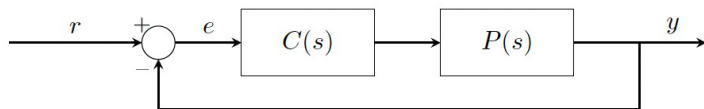


Figure 3: Standard feedback architecture.

In order to design the compensator, the problem has been split into the following subtasks:

3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45° .

- a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system. $\omega_c = 10^2$
- e) Finally, by relying on the definition of the crossover frequency, determine a suitable value for the gain K of the compensator.

Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5 \frac{(1 - \frac{s}{200})}{(1 + \frac{s}{20})(1 + \frac{s}{4})}$$

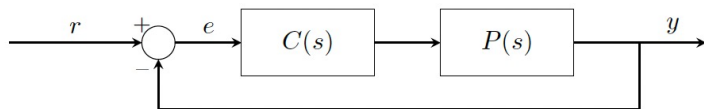


Figure 3: Standard feedback architecture.

$$C_{lead} = \frac{s/a + 1}{s/b + 1}, 0 < a < b$$

In order to design the compensator, the problem has been split into the following subtasks:

3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45° .

$$\omega_c = 10^2$$

- a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.
- e) Finally, by relying on the definition of the crossover frequency, determine a suitable value for the gain K of the compensator. $K = 6.56 \quad a \approx 29 \text{ \& } b = 348$

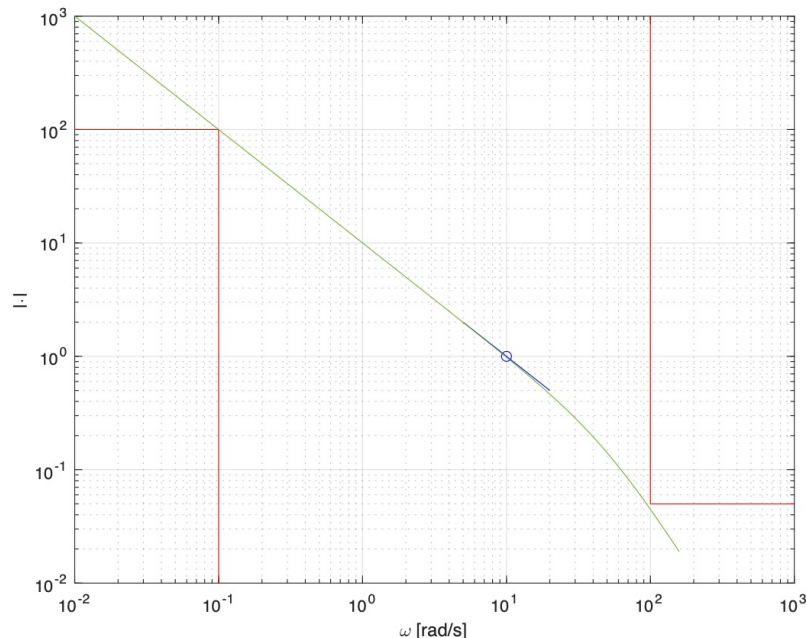
putting everything together we get:

$$C_{lead}(s) = 6.56 \frac{1 + s/29}{1 + s/348}$$



Performance Limitations

- The requirement to keep a shallow slope over a “large” interval near crossover may make it impossible to clear the obstacles
- nmn-phase zeros and unstable OL poles may make the situation even harder
- Many performance goals conflict, e.g., you cannot achieve strong disturbance rejection, noise attenuation, and good tracking at the same time



Tips for Problem Set 10

Easy	Medium	Hard
	1, 2, 3	4

1: a) look up the graphical definitions in the slides. c) think about negative & positive margins. d) recall the Nyquist stability criterion

2: a) $|kL(j\omega_{gc})| = 1$

b) look at the mathematical definitions for phase & gain margin and plug in the open-loop TF

3: a) we did it graphically together for intuition, try to do the calculations if you wanna get the exact results b) follow the tips and remember $e_{ss} = \lim_{s \rightarrow 0} \frac{1}{1+L(s)}$

for small ω $L(s)$ large, so $\lim_{s \rightarrow 0} \frac{1}{1+L(s)} = \frac{1}{L(s)}$

which compensator do we choose to increase magnitude?

4.1: we need atleast as many poles @ origin, as order of ramp b) remember the table summarizing e_{ss} or use FVT

4.2: use the definition of sensitivity function to calculate the error

4.3: remember how we changed ω_{gc} in our example by multiplying the TF with a gain K

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?
I would appreciate your feedback. Please let me know.

<https://docs.google.com/forms/d/e/1FAIpQLSdHI0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header>